

Incremental forming

Incremental forming is a flexible process that shapes sheets of various materials using plastic deformation with a tool to achieve the desired shape. Unlike traditional manufacturing processes, it does not require dies or molds, making it ideal for low-volume production, prototyping, or custom parts. The tool follows a programmed path to manufacture thin shell shapes, offering high flexibility and reduced setup and production costs. It also enables the production of various products in small quantities, where mold creation would be too expensive.

An example of incremental forming in use today is by car manufacturers like Ford, Nissan, and Lexus, who use it to prototype new vehicle designs and add minor customizations to vehicles, such as the custom add-on on the Lexus LS edge, as highlighted in the red circle of the figure referenced.

Previous Research

Previous research developed a simple CAM system to generate a tool path and conducted experiments to test it. The experimental conditions will be described later. The CAM system, developed using Python, can be described in four steps:

First, STL data from 3D CAD is converted into a point cloud, a collection of points derived from the triangles in the STL file. Next, the point cloud's points are sorted into contour lines or layers defined by the Z coordinate, identifying points on the outermost part of the shape to ensure the toolpath avoids the middle of the forming shape. The third step

constructs the toolpath by calculating distances from the tool's current position to other points on the layer, moving to the nearest point, and repeating this for all points on each layer. Finally, the X, Y, and Z coordinates are converted into G-code for machine readability.

Research Progress (CAM system changes)

Based on previous research, the CAM system has been enhanced to handle complex shapes. Additions include initial and ending commands for toolpaths, convex hull processing, reverse offset, and other features to streamline the transition from design to machining.

In the updated CAM system, STL files are processed by rounding Z coordinates to the nearest whole number, forming a point cloud with a uniform layer distance of 1 mm, and processing from the highest to the lowest layer. The system uses the convex hull library to find the outer boundary of points, removing layers with fewer than 5 points. Points within a certain radius of the current point are filtered, and the toolpath is adjusted for tool size and spring back. The points are re-sorted for a smooth toolpath, ensuring all points are visited once, and each layer is closed by connecting the last point to the first.

The generated toolpath is lowered by 50.5 mm to adjust for the workpiece origin, with X, Y, and Z coordinates rounded for accuracy. Finally, the coordinates are converted to G-code, incorporating syntax, initial setup commands (e.g., spindle speed, feed rate), and a termination sequence that moves the tool

to a safe height and stops the machine. These changes enable the CAM system to generate smooth toolpaths for complex shapes, optimize the process, and allow immediate machining without adjustments.

Research Progress (Test shape changes)

Since the last interim presentation, the test shape has been revised due to sharp edges and issues with small areas inaccessible to the tool because of its diameter. The updated shape is a simpler, shorter version, with dimensions of 30 mm x 30 mm x 9.9 mm, designed to avoid 90-degree angles while retaining some complexity.

Research progress (Experiments/analysis)

Experiments were conducted using an MC430L machining center on 110 mm x 110 mm x 1 mm nylon sheets, with a mold of 89.92 mm x 89.92 mm x 20 mm for the 30 mm x 30 mm x 9.9 mm test shape. The tool was a 2.5 mm rod attached to a MEGA Micro type S Chuck.

The mold successfully formed the test shape, though some areas were thinner due to the toolpath being too close to the mold. Experiments with the mold yielded better results, as the tool pressed the material against the mold during formation. Tests used a feed rate of 200–400 mm/min and a spindle speed of 250–300 RPM.

Without the mold, the material was thicker post-machining, but could not reach the desired depth due to spring back. Lowering the tool's origin resolved this. These tests used a feed rate of 200–

1000 mm/min and a spindle speed of 250–500 RPM. Forming without the mold was more challenging due to higher spring back, requiring higher feed rates and spindle speeds to maintain the shape.

Future goals

Between now and the next presentation, the goal is to conduct experiments with CFRTP in an oil bath, with and without mold, and to learn to use a 3D scanner for in-depth analysis of the machined shape, as it is challenging to do so right now. After successfully forming the test shape and mastering the 3D scanner, the focus will be on improving formability and analyzing characteristics, time permitting.

The experimental setup will match previous research, using a T-22H oil bath and silicon oil to heat the CFRTP. The test piece is a 110 mm x 110 mm x 1 mm sheet of discontinuous short fiber CFRTP. Experimental conditions will adopt the feed rate, spindle rotation, pitch, and temperature from the previous research's best-performing experiment, while radius, angle, and depth will align with the updated test shape's parameters